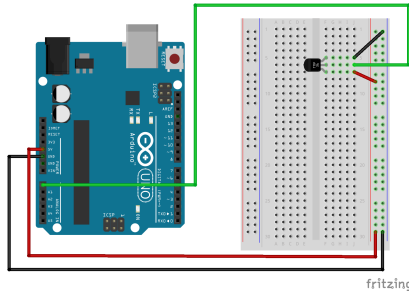


Element G

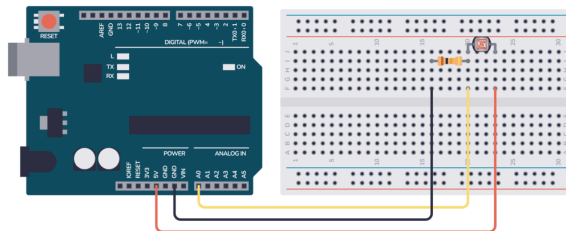
Arduino

At first we built separate prototypes on the arduino. We had two separate arduinos for each part. We build one for light intensity, and one for temperature.

Initial prototypes:

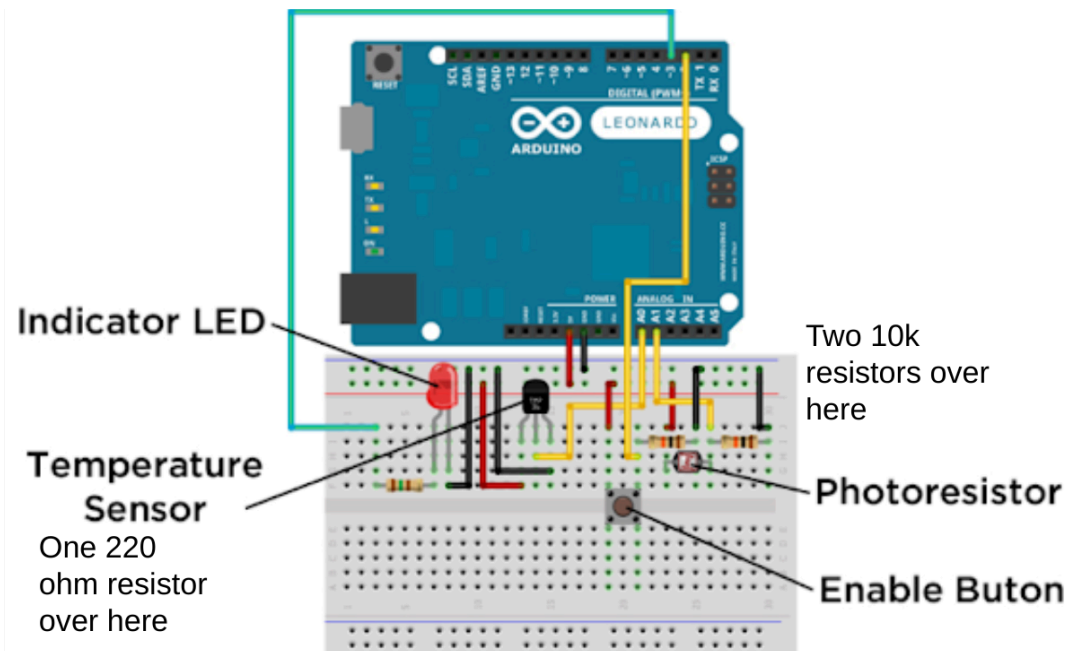


This is our temperature design for the arduino.



This is our first light intensity circuit. This worked individually but we had to find a way to combine the circuits.

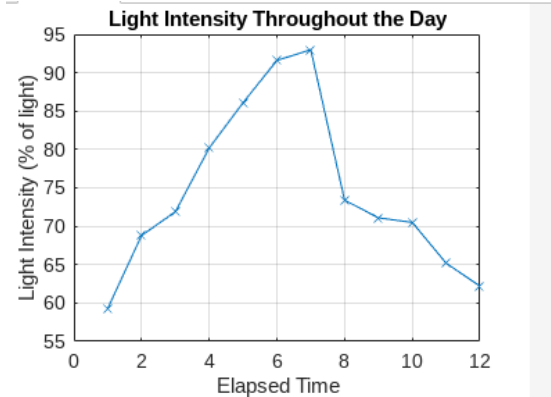
Our final working Arduino prototype:



This combines the temperature sensor and light intensity sensor. It starts with the click of a button. Once started the arduino put the data into a csv file and we took the data from the csv file then imported it into matlab.

We tried to add the humidity sensor into the arduino but we couldn't get it working. As we tried to change the arduino code we were unable to get data out of the sensor. In order to get the light intensity, temperature, and humidity into one csv file we had to have them in the same circuit. We were unsuccessful and ran out of time when trying to add the humidity sensor we couldn't so we couldn't get the data into one file to then import into matlab.

Design Requirements	Successfulness
Get correct and accurate measurements for temperature, light intensity, and humidity.	The light intensity data is accurately measured. However, we were not able to implement a humidity sensor, so we had no data at all for humidity. In addition, since we don't know the scale the temperature sensor uses, we cannot know for sure if the temperature data is accurate.

Graphical Representation	 We were very successful creating a light intensity graph in matlab from the data we collected. The temperature sensor was non linear and we were unable to figure out how the amount of volts the arduino read and the temperature in fahrenheit were correlated.
Set cutoff data requirements	We were supposed to set cutoff points for temperature (85 degrees F) and humidity (95%). Since we were unable to find the correct values for temperature because the voltage reading was not linear. This made us unable to figure out fahrenheit values from the voltage reading. Since we were unable to add the humidity sensor we did not have the humidity sensor cutoff at 95% humidity.
Retrieve light intensity data every 15 minutes	No, we got it every 2,000 seconds.
Retrieve humidity and temperature data every 20,000 seconds	No, we got temperature data every 2,000 seconds, and we weren't able to get humidity data due to the lack of a sensor.
Collect continuous data for 3 days	Due to school hour restrictions, we instead had 3 separate sets of data for three separate school days.
Small and compact	This device was small and compact. It fit on the bookshelf and did not take up too much space.
Positioning of the device	Successful, it was near an outlet, and didn't require extension cords.
Less than 40 lines of MatLab code	This was successful, there were only 35 lines of code.

Since the light sensor had a range from 0-1023 we had to map it from 0-100%. I decided to map the light with $x/1023$. X would be the actual light intensity value the light sensor read in decimal form.

When trying to access matlab code to copy it into this document there has been a error with matlab loading.